

- Let $x = a, b, c$. Define relation **J** on $P(x)$ as follows: $\forall A, B \in P(x), A\mathbf{J}B \Leftrightarrow A \cap B \neq \emptyset$
- Is $\{A\}\mathbf{J}\{C\}$? No, because $\{A\} \cap \{C\} = \emptyset$
 - Is $\{a, b\}\mathbf{J}\{b, c\}$? Yes, because $\{a, b\} \cap \{b, c\} = \{b\} \neq \emptyset$
 - Is $\{a, b\}\mathbf{J}\{a, b, c\}$? Yes, because $\{a, b\} \cap \{a, b, c\} = \{a, b\} \neq \emptyset$

- Let $A = \{3, 4, 5\}, B = \{4, 5, 6\}$ and let **S** be the “Divides” relation: $\forall (x, y) \in A \times B, x\mathbf{S}y \Leftrightarrow x|y$
- $S = \{(3, 6), (4, 4), (5, 5)\}$
 - $S^{-1} = \{(6, 3), (4, 4), (5, 5)\}$

Suppose a function $F : X \rightarrow Y$ is onto but not one-to-one. Is F^{-1} a function? No, because if $F : X \rightarrow Y$ is not one-to-one, then $\exists (x_1, x_2) \in X \wedge \exists y \in Y, x_1 \neq x_2 \wedge (x_1, y) \in F \wedge (x_2, y) \in F$. But this implies $\exists x_1, x_2 \in X \wedge \exists y \in Y, x_1 \neq x_2 \wedge (y, x_1) \in F^{-1} \wedge (y, x_2) \in F^{-1}$. Consequently, F^{-1} does not satisfy property (2) of the definition of function.

Define relations **R** and **S** on $\mathbb{R}$ as follows:
R = $\{x, y\} \in \mathbb{R} \times \mathbb{R} | x^2 + y^2 = 4$ Square!
S = $\{x, y\} \in \mathbb{R} \times \mathbb{R} | x = y$ Line!
 Graph $R, S, R \cup S, R \cap S$ in the Cartesian plane. *R, Circle. S, Line. R and S, both of them. R or S, just the intersection points.*

- Let $A = \{2, 3, 4, 6, 7, 9\}$ and define a relation **R** on A as follows: $\forall x, y \in A, x\mathbf{R}y \Leftrightarrow 3|(x - y)$.
- Each element has a loop to itself.
 - If element a is related to element b , then b is also related to a .
 - In each case where there is an arrow from one point to another point, and that to a third, there is an arrow going from the first to the third.

- 1. Reflexive? Yes.**
- 2. Symmetric? HELL YES.**
- 3. Transitive? HELL TO THE HELL TO THE YES**

R is reflexive iff $\forall x \in A, x\mathbf{R}x$ **R** is symmetric iff $\forall x, y \in A, x\mathbf{R}y \rightarrow y\mathbf{R}x$ **R** is transitive iff $\forall x, y, z \in A, x\mathbf{R}y \wedge y\mathbf{R}z \rightarrow x\mathbf{R}z$

Let $A = \{0, 1, 2, 3\}$ and define relations **R, S, T** as follows:
R: $\{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$
S: $\{(0, 0), (0, 2), (0, 3), (2, 3)\}$
T: $\{(0, 1), (2, 3)\}$

- Is **R** reflexive, symmetric, transitive? Yes, Yes, Nah.
- Is **S** reflexive, symmetric, transitive? No, no, yasss.
- Is **T** reflexive, symmetric, transitive? No, no, yasss.